

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.  
(2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.  
(3) Answers to the **same question** must be **written together**.  
(4) Numbers to the **right** indicate **marks**.  
(5) Draw **neat labeled diagrams** wherever **necessary**.  
(6) Use of **Non-programmable** calculator is **allowed**.

**1. Attempt any two of the following:** **12**

- a. Explain the role of application frameworks in ASP.NET Core. How do they help in building web applications?
- b. Discuss the steps to create and run an ASP.NET Core project using a code editor of your choice.
- c. Describe the process of creating your first ASP.NET Core application. What are the initial configurations needed?
- d. Explain the importance of unit testing in ASP.NET Core applications. How can developers create and run unit tests effectively?

**2. Attempt any two of the following:** **12**

- a. Explain how to add data and display a list of products in an ASP.NET Core application. How does pagination improve the user experience?
- b. What are the benefits of building a shopping cart in ASP.NET Core with robust navigation controls? Provide a detailed explanation of how this can be achieved.
- c. Discuss how a service can refine the cart model in ASP.NET Core. How does this refinement enhance the functionality and user interaction with the cart?
- d. Describe the steps involved in preparing Blazor Server for administration purposes. How can it be utilized to manage orders and maintain catalog integrity?

**3. Attempt any two of the following:** **12**

- a. Explain the process of creating the Identity Database in ASP.NET Core. What are the key components of identity management?
- b. Describe how to add a conventional administration feature in an ASP.NET Core application. What are the benefits of having such a feature?
- c. Discuss the steps involved in applying a basic authorization policy. How can you test the effectiveness of this security policy?
- d. What are the essential preparations needed for deploying an ASP.NET Core application? Outline the steps involved in the deployment process.

**4. Attempt any two of the following:** **12**

- a. Explain the role of caching in enhancing the performance of ASP.NET Core applications. How can developers implement caching strategies effectively?
- b. Discuss how to create and run a RESTful web service using both the Minimal API and a controller. Compare the two approaches in terms of complexity and usability.
- c. What are the challenges associated with managing related data in ASP.NET Core? How can developers ensure efficient data retrieval?

- d. Describe the significance of configuring the request pipeline correctly in an ASP.NET Core application. What common middleware components should be included?

**5. Attempt any two of the following:**

**12**

- a. Explain the concept of views in ASP.NET Core MVC. How do you modify a Razor view to enhance its functionality?
  - b. Discuss the Razor syntax and its role in creating dynamic web pages. Provide examples of common Razor syntax usage.
  - c. What is the View Bag in ASP.NET Core, and how is it used to pass data from controllers to views? Explain with an example.
  - d. Describe the use of Temp Data in ASP.NET Core applications. In what scenarios is Temp Data particularly useful?
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